

Weekly Problem 4

Ravi Kini

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Exercise 1

Part (a)

1	$\neg p \vee q$	
2	p	
3	$\neg p$	
4	$\neg p \vee (p \wedge q)$	$\vee I3$
5	q	
6	$p \wedge q$	$\wedge I2, 5$
7	$\neg p \vee (p \wedge q)$	$\vee I6$
8	$\neg p \vee (p \wedge q)$	$\vee E1, 2-4, 5-7$
9	$\neg p$	
10	$\neg p \vee (p \wedge q)$	$\vee E9$
11	$\neg p \vee (p \wedge q)$	$TND2-8, 9-10$
12	$p \Rightarrow (p \wedge q)$	$mcI11$

We then see that $\neg p \vee q \vdash p \Rightarrow (p \wedge q)$ is a valid argument in the Logic of Paradox.

Part (b)

Let $p = i$, $q = 1$, and $r = 0$. We then see that $p \wedge q = i$, so $(p \wedge q) \Rightarrow r = i$. However, $q \Rightarrow r = 0$. Consequently, the argument $p, (p \wedge q) \Rightarrow r \vdash q \Rightarrow r$ is an invalid argument in the Logic of Paradox.